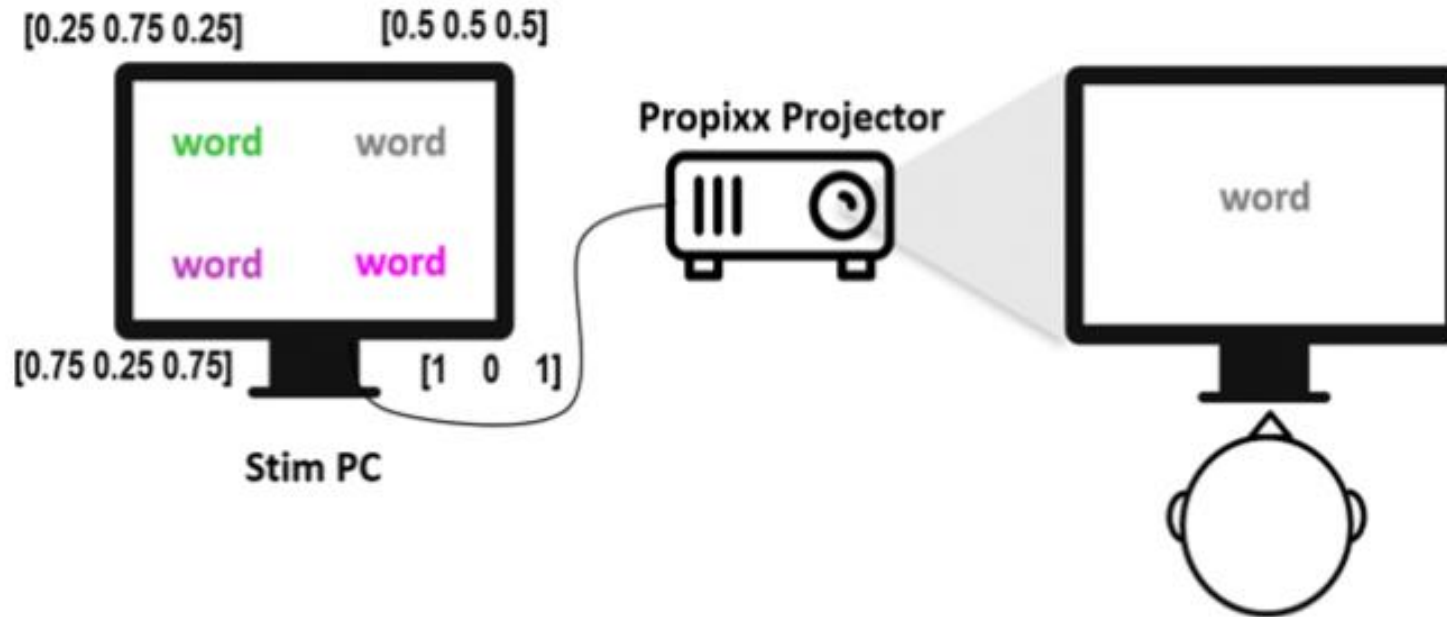


RIFT tutorial

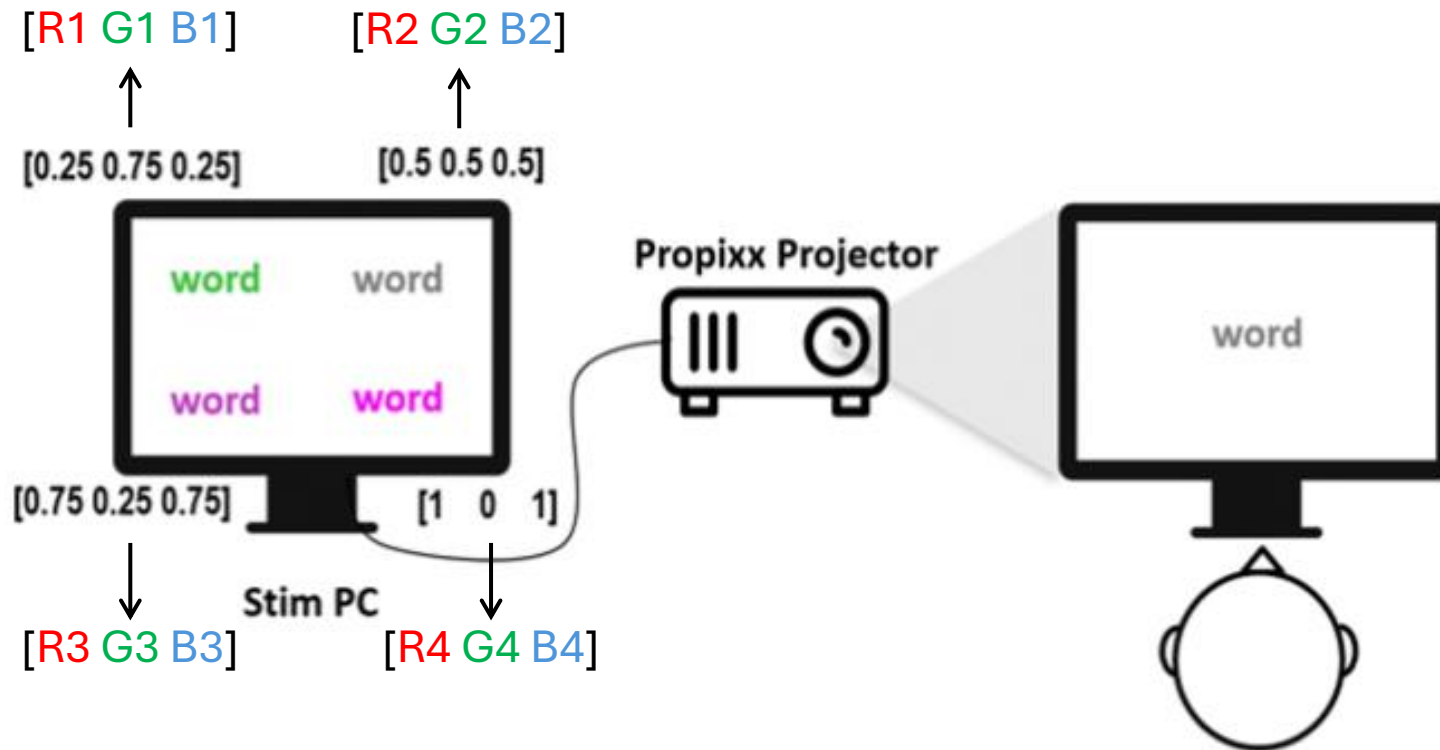
Yali Pan

yalipan666@gmail.com

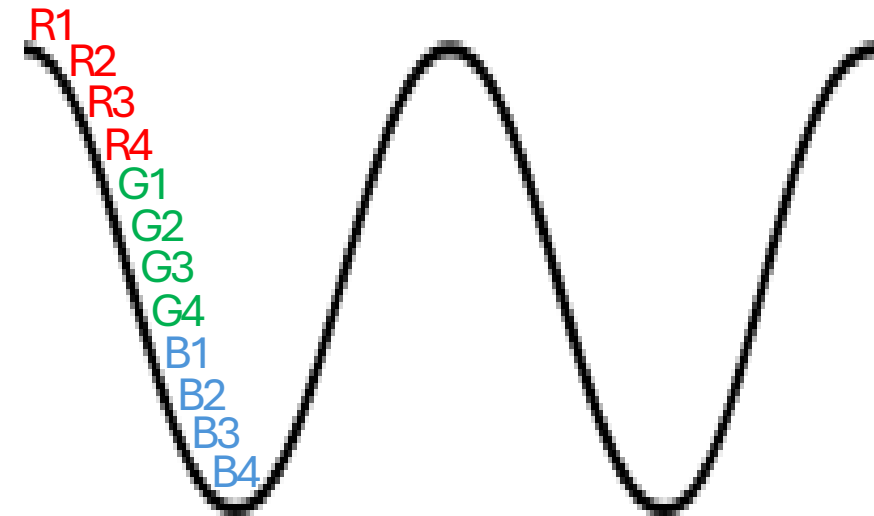
How PROPixx works in the rapid mode?



- Refresh Rate(Hz): Stim PC 120; Projector 1440
- Projector takes color parameters in Stim PC as luminance
- The fast flicker is invisible and drives neurons in the visual system at the same frequency



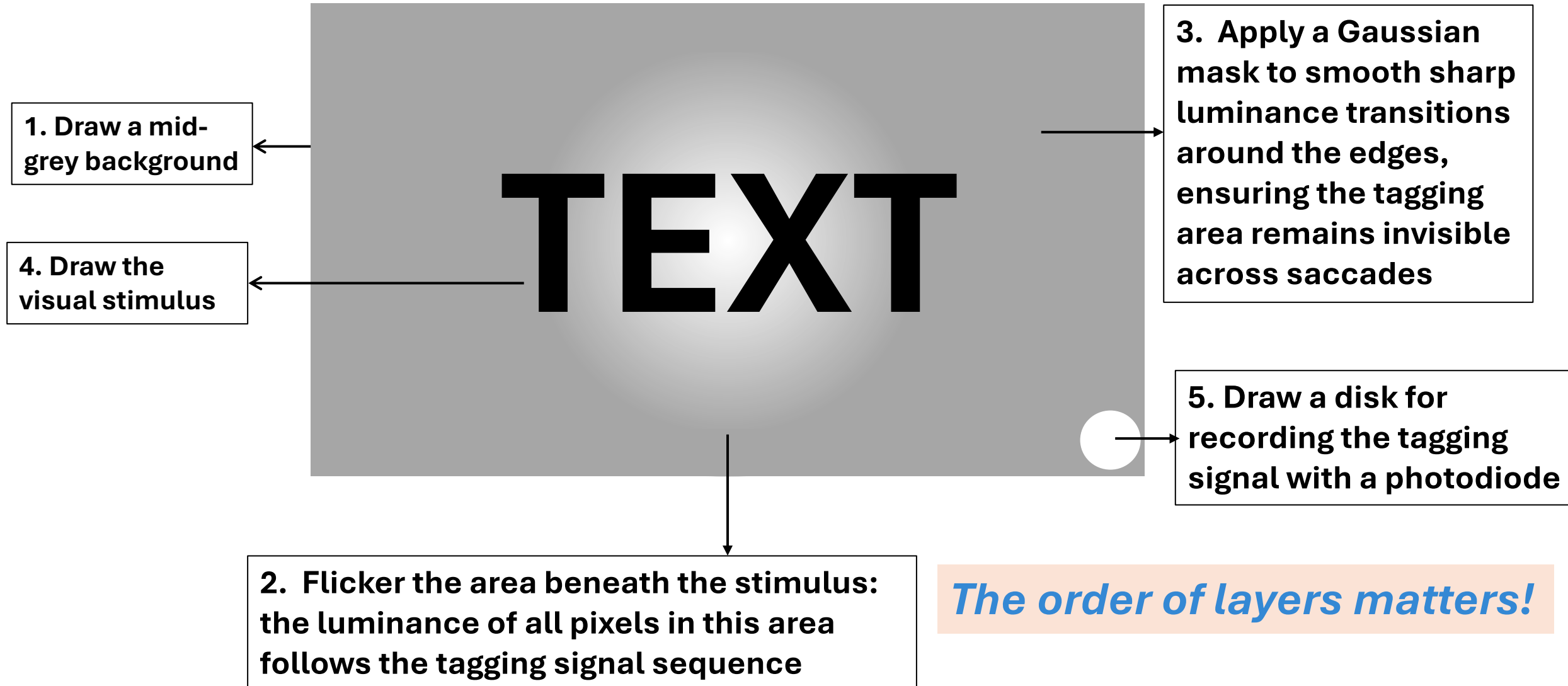
Tagging luminance sequence



- Refresh Rate(Hz): Stim PC 120; Projector 1440
- Projector takes color parameters in Stim PC as luminance
- The fast flicker is invisible and drives neurons in the visual system at the same frequency

- **Colour codes for Q1 in the 1st frame: luminance([1,5,9])**

How to frequency tag a (visual) stimulus?



RIFT requires very rapid stimulus presentation

- Use “**offscreen**” to pre-draw all stimuli, e.g., the sentence, the flickering patch, the gaussian mask (to prevent the presentation from slowing down over time)
- Use “frame loops” to control the timing of the presentation
- Pre-draw all stimuli outside of the loop, only change the colour of the flickering patch within the presentation loop
- Clear all offscreen after each trial to clear buffers

What are the steps of data analysis?

S1_Get_ExpInfo.m

Bad sensors, triggers, experimental design

Get_MEGData.m

Data, header, MEG triggers

ft_preprocessing.m

Remove bad sensors, bandpass filter ([0.5 100]Hz) & notch filter ([49 51]Hz)

S2_PreProcessing.m

ICA4rawdata.m

Artefact rejection by ICA (oculomotor & heartbeat)

Get_EyeData.m

Extract EM events and triggers from eye link file, “locate” EM events with sentence ID & word ID

Get_Event.m

Align MEG triggers and EM triggers, create events using experiment design and EM events (wrt MEG)

Event.event_raw_header

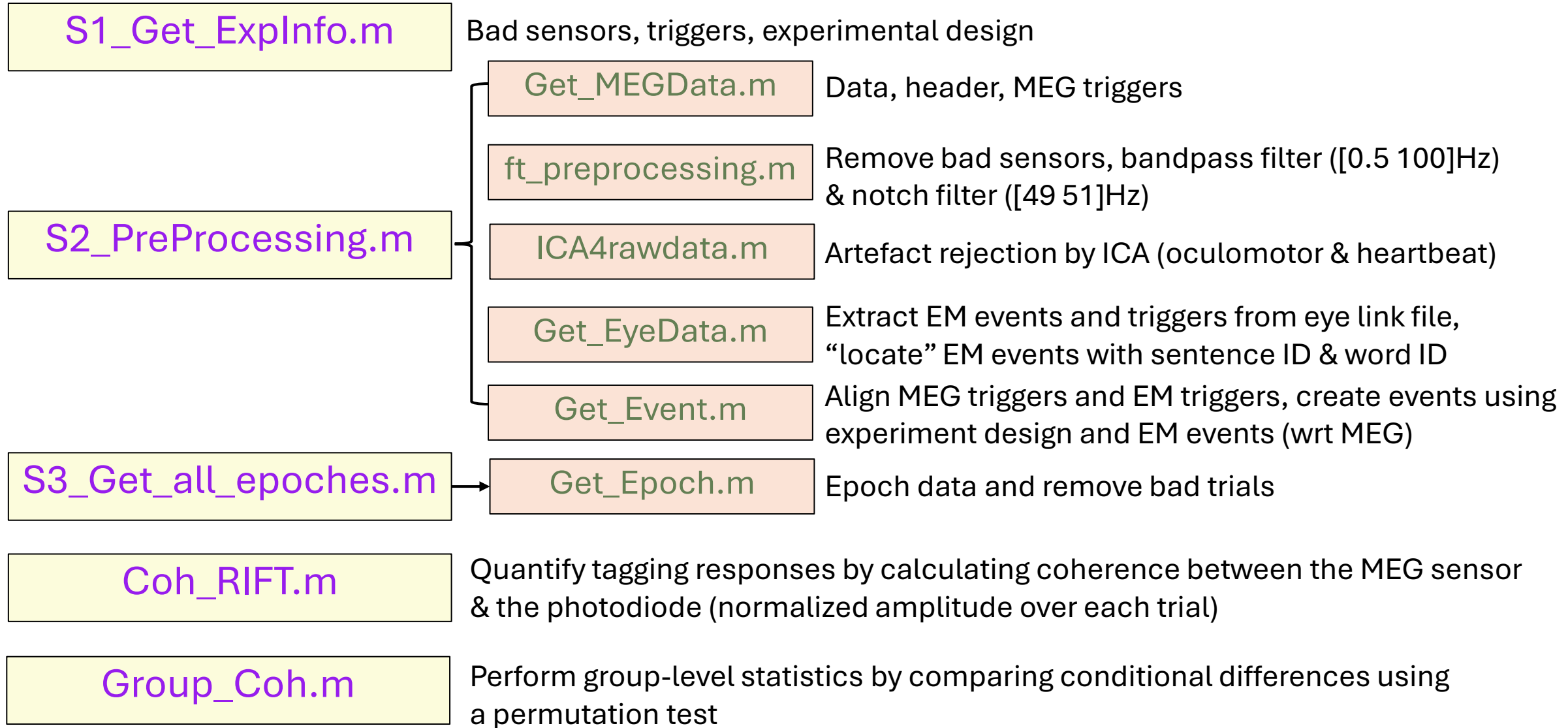
	1	2	3	4	5	6	7	8	9	10
1	sentence_id	word_loc	loc2targ	SentenceCondition	fixation_on_MEG	saccade2this_duration	fixation_duration	NextOrder	FirstPassFix	PreviousOrder

Experiment design

Triggers to
epoch MEG data

Eye movement measures to select events

What are the steps of data analysis?



How to measure time-resolved tagging response?

1. Select the tagging response sensors

- Tagging response: coherence between a MEG sensor and the photodiode at the tagging frequency band
- Loop over all sensors (in the occipital cortex), over all trials (regardless of conditions)
- Compare the coherence during the time-of-interest (e.g., the pre-target fixation interval) and the baseline (i.e., no flicker interval)

- `ft_statfun_indepsamplesZcoh.m`

(one tail, alpha=0.01, 2000 permutations)

$$Z = \frac{(\tanh^{-1}(|\text{coh}_1|) - \text{bias}_1) - (\tanh^{-1}(|\text{coh}_2|) - \text{bias}_2)}{\sqrt{\text{bias}_1 + \text{bias}_2}}$$

$$\text{bias}_1 = \frac{1}{2n_1 - 2}, \text{bias}_2 = \frac{1}{2n_2 - 2}$$

where coh_1 and coh_2 denote the coherence value for the pre-target and baseline segments, bias_1 , and bias_2 is the term used to correct for the bias from trial numbers of the pre-target (n_1) and baseline condition (n_2). All trials from the pre-target and baseline conditions were used.

How to measure time-resolved tagging response?

1. Select the tagging response sensors
2. Obtain the time-frequency plot of coherence (only for tagging response sensors)
 - Loop over conditions (coherence is calculated over trials)
 - Loop over frequency-of-interest F_i (e.g., [40:1:80]Hz)
 - Obtain **Hilbert** analytic signals of the filtered data ($F_i \pm 5\text{Hz}$), [MEG + photodiode]
 - calculate coherence between each tagging response sensor and the photodiode, then average coherence over sensors

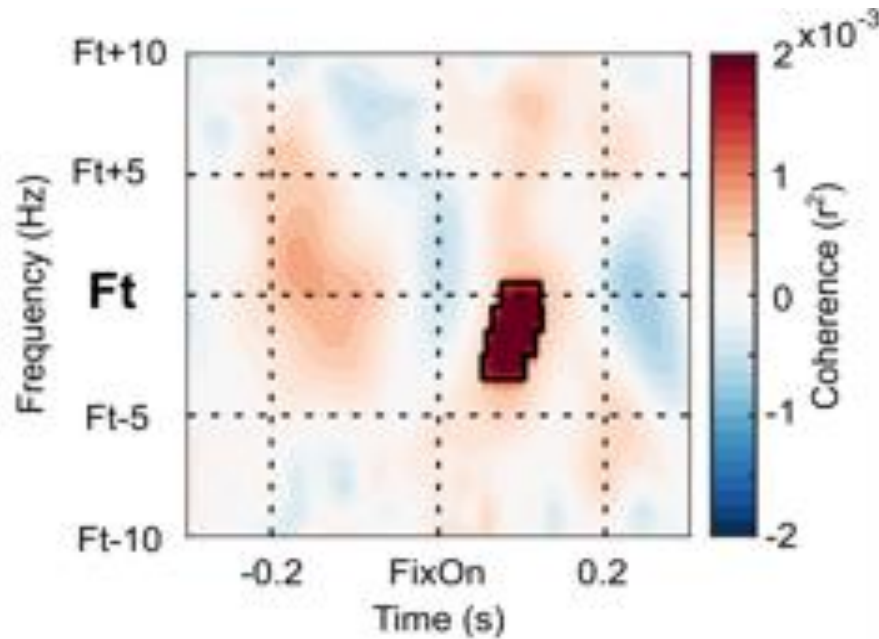
$$\text{coh}(t) = \frac{\left| \frac{1}{n} \sum_{j=1}^n m_{x_j}(t) m_{y_j}(t) e^{i\phi_{xy_j}(t)} \right|^2}{\sum_{j=1}^n m_{x_j}(t)^2 m_{y_j}(t)^2}$$

where n is the number of trials. For the time point t in the trial j , $m_x(t)$ and $m_y(t)$ are the time-varying magnitude of the analytic signals from a MEG sensor (x) and the photodiode (y) respectively, $\phi_{xy}(t)$ is the phase difference as a function of time (for a detailed description, please see [Cohen, 2014](#)).

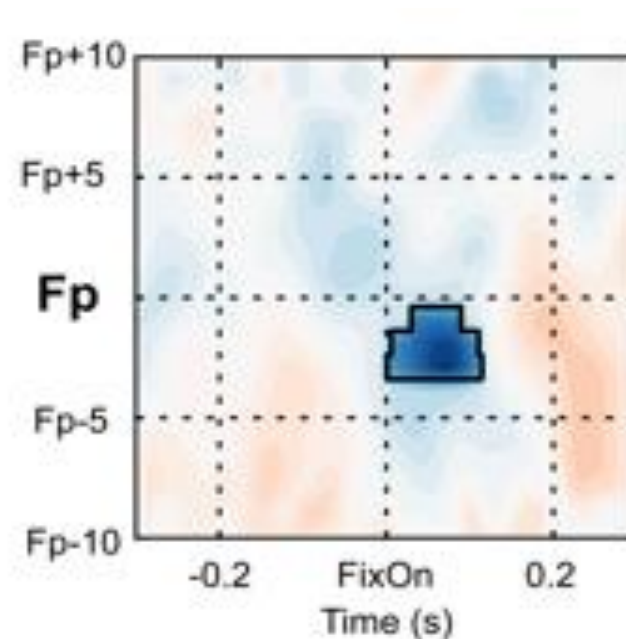
How to do statistics (on group level)?

Non-parametric permutation test, conditional contrast

Pre-target interval, target tag@Ft



Target interval, post-target tag@Fp



(Pan, Frisson, Snell, Federmeier, Jensen, in prep.)

- Doesn't require a "manual" selection of the time and frequency window, especially convenient when tagging more than one stimulus simultaneously.